MediQueue – Non-Functional Requirements

1. Document Control

Project Name: MediQueue – Clinic Appointment & Queue Management System
Document Title: Non-Functional Requirements
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Prepared By: Project Coordinator
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2. Requirement Rating Guide

Priority	Meaning	How to use
Must	Essential for a safe, usable baseline.	Treat as required for a proposed release.
Should	Important but can be negotiated if constraints require it.	Document any agreed exception.
Could	Useful enhancement.	Consider after core requirements.

3. Non-Functional Requirements

ID	Quality Area	Priority	Requirement	Verification / Acceptance
NFR-01	Security	Must	Only authenticated users may access protected functions; public pages must not expose private patient data.	Test unauthenticated access to protected pages and APIs; access is denied.
NFR-02	Role-based access	Must	Patient, Receptionist/Staff and Administrator permissions must be separated.	Attempt restricted actions with each role; unauthorized actions are blocked.

NFR-03	Password protection	Must	Passwords must never be stored or displayed as plain text; use a suitable salted password-hashing approach if authentication is implemented.	Inspect stored test accounts and confirm plain-text passwords are absent.
NFR-04	Session/token handling	Must	Authentication tokens or sessions must be handled securely and invalidated on sign-out where supported.	Sign-out prevents subsequent protected access using the ended session/token.
NFR-05	Data validation	Must	Validate required fields, formats, lengths and allowed values on both client and server where applicable.	Invalid or missing input is rejected with a clear validation message.
NFR-06	Privacy	Must	Collect only information needed for appointment and queue workflows; restrict access to personal data.	Review data fields and role permissions; no unnecessary sensitive data is requested.
NFR-07	Performance	Should	For a small demo dataset and local/test environment, common pages should normally respond within 3 seconds under ordinary use.	Record response times for representative operations and document environment/dataset.
NFR-08	Concurrent use	Should	The system should handle a small clinic team and multiple patients without duplicate queue numbers or conflicting appointment status.	Run basic concurrent/manual tests and check for duplicates/conflicting updates.
NFR-09	Usability	Must	A first-time user should be able to find appointment booking, appointment status and queue status using clear labels.	Conduct a simple task walkthrough; record confusing labels and fixes.
NFR-10	Accessibility	Should	Use readable contrast, labels, keyboard-operable controls and meaningful error messages.	Review main forms and navigation with a basic accessibility checklist.
NFR-11	Responsive layout	Should	Core pages should remain usable on common desktop, tablet and mobile viewport sizes.	Check representative screen widths and record layout defects.

NFR-12	Reliability	Should	Failed submissions should show an error and should not silently create duplicate appointments or queue entries.	Repeat submissions and simulate invalid input; check persisted records.
NFR-13	Data integrity	Must	Appointment and queue status values must follow defined valid states and transitions.	Try invalid status values and verify rejection or controlled transition.
NFR-14	Availability	Could	For a portfolio demo, the system is available while the local/test server is running; production uptime is not promised.	Document server/environment limits; no production SLA is claimed.
NFR-15	Compatibility	Should	Support current versions of major browsers used for testing, subject to the project environment.	Run a basic smoke test in available browsers and record versions.
NFR-16	Maintainability	Should	Use consistent naming, modular components, and documented setup/run instructions.	Review repository structure and setup documentation.
NFR-17	Logging and errors	Should	Show user-friendly errors and keep useful diagnostic details in development logs without exposing secrets.	Review failure scenarios and confirm secrets/tokens are not logged.
NFR-18	Backup/recovery	Could	Provide a documented way to export or back up demo data before major changes if supported by the chosen database.	Demonstrate or document a test backup and restore procedure.
NFR-19	Notifications	Should	If notification functionality is included, the system must clearly indicate whether a notice was queued/sent; external delivery is not guaranteed without integration.	Test internal notification records; label external delivery as out of scope unless integrated.
NFR-20	Documentation	Must	Provide concise user/administrator guidance and project setup notes for the portfolio handover.	Reviewer can follow the instructions in a fresh test setup.

4. Security and Privacy Checklist

Check ID	Check	Evidence to record	Status
SEC-01	Protected pages/APIs require sign-in	Test result or screenshot	Not tested
SEC-02	Roles cannot perform each other's restricted actions	Role-permission test table	Not tested
SEC-03	Passwords are not stored in plain text	Code/configuration and test data review	Not tested
SEC-04	Inputs are validated on server where applicable	Validation test cases	Not tested
SEC-05	No passwords, secrets or tokens in logs/repository	Repository/config review	Not tested
SEC-06	Personal data is visible only to permitted roles	Access test cases	Not tested

5. Assumptions and Constraints

Type	Item	Details
Assumption	A-01	Initial use is a student portfolio/demo; production deployment is not assumed.
Assumption	A-02	Performance targets are preliminary and must be tested in a stated environment.
Constraint	C-01	Available time, team capacity, hardware and free-tier tools may limit testing.
Constraint	C-02	SMS/email delivery and third-party integrations are not guaranteed unless separately implemented and configured.

6. Review and Change Control

Review each requirement with the project scope and functional requirements. Record proposed changes, rationale, impact on time/cost/risk, decision, owner and date. Do not mark a requirement as tested or approved until evidence exists.

Change ID	Requirement ID	Requested change	Impact	Decision	Owner / Date
CR-01	TBD	TBD	Assess schedule, scope, risk and dependencies	Pending review	TBD